

mixmogam 2.0: Technical Details of the Mixed-Model Association Engine

mixmogam contributors

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1 Overview and notation

mixmogam fits the standard linear mixed model for genome-wide association mapping,

$$\mathbf{y} = \mathbf{X}\beta + u + e, \quad u \sim \mathcal{N}(0, \sigma_g^2 \mathbf{K}), \quad e \sim \mathcal{N}(0, \sigma_e^2 I_n), \quad (1)$$

where n is the number of samples, $\mathbf{X} \in \mathbb{R}^{n \times q}$ the covariate design (an intercept is included by default), and $\mathbf{K} \in \mathbb{R}^{n \times n}$ a kinship (relationship) matrix. We write $\delta = \sigma_e^2 / \sigma_g^2$ and $\mathbf{V} = \sigma_g^2 (\mathbf{K} + \delta I_n)$. The covariance of (1) is parameterized by the single ratio δ ; all variance-component estimation reduces to one-dimensional optimization of the REML or ML profile likelihood over δ .

The package implements four regimes:

1. **Exact** ($n \lesssim 5,000$ by default): one dense eigendecomposition per leave-one-chromosome-out (LOCO) kinship; the association scan evaluates the exact EMMAX statistic in batched dense GEMMs (Sections 2, 3, 7).
2. **Two-step K-free** (large n): BOLT-LMM and LDAK-KVIK statistics with LOCO residuals from conjugate gradients or variational Bayes, never forming an $n \times n$ matrix (Section 7), with an optional structure-aware denominator (Section 7.4).
3. **SLQ fitting** (large n): variance components by deflated stochastic Lanczos quadrature, requiring only matrix-vector products with $\mathbf{K}$ and no $O(n^3)$ factorization (Section 6).
4. **Truncated spectrum**: a randomized top- k eigenbasis of $\mathbf{K}$ with a flat-bulk tail (Section 5).

Kept for comparison only: its scans were anti-conservative in simulation, and they warn.

Genotypes are stored sample-major as $n \times m$ matrices of `int8` dosages in $\{0, 1, 2\}$ with -1 encoding missing calls; SNP blocks are converted to `float32` with fused mean imputation immediately before use, so the float working set is bounded by the block size regardless of m .

2 Variance-component estimation: exact EMMA

2.1 Restricted eigenspace

Following Kang et al. (2008), the REML likelihood can be expressed through the spectral decomposition of the *restricted* matrix $\mathbf{M} = \mathbf{S}(\mathbf{K} + I_n)\mathbf{S}$ with $\mathbf{S} = I_n - \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ (the v1 construction; eigenpairs beyond the q annihilated directions, offset by one). mixmogam evaluates the identical likelihood from a *single* eigendecomposition of $\mathbf{K}$ via the matrix determinant lemma: with $V = \mathbf{K} + \delta I_n = \mathbf{U} \text{diag}(\lambda_i + \delta) \mathbf{U}^\top$ and $W = \mathbf{U}^\top \mathbf{X}$, $w = \mathbf{U}^\top \mathbf{y}$,

$$\text{RSS}(\delta) = w^\top D w - b^\top A^{-1} b, \quad A = W^\top D W, \quad b = W^\top D w, \quad D = \text{diag}\left(\frac{1}{\lambda_i + \delta}\right), \quad (2)$$

Algorithm 1 EMMA exact fit (REML shown; ML analogous)

Require: $\mathbf{y}$, $\mathbf{X}$, $\mathbf{K}$, grid $(G, l_{\min}, l_{\max})$, tolerance τ

- 1: $\mathbf{Q} \leftarrow$ thin QR of $\mathbf{X}$; $\mathbf{S} \leftarrow I - \mathbf{Q}\mathbf{Q}^\top$
 - 2: eigendecompose $\mathbf{M} = \mathbf{S}(\mathbf{K} + I)\mathbf{S}$; keep $\{\lambda_i^R, \eta_i\}_{i \leq p}$
 - 3: vectorize $\ell_R(\delta_j)$, $\ell'_R(\delta_j)$ over the grid $j = 0..G$
 - 4: $i^* \leftarrow \arg \max_j \ell_R(\delta_j)$
 - 5: find i with $\ell'_R(\delta_i) > 0 > \ell'_R(\delta_{i+1})$ maximizing mean endpoint likelihood
 - 6: $\hat{\delta} \leftarrow \text{Brent}(\ell'_R, \delta_i, \delta_{i+1}, \tau)$; accept per boundary rules, else fall back to δ_{i^*}
 - 7: **return** $\hat{\delta}$, $\hat{\sigma}_g^2 = s_1(\hat{\delta})/p$, h_{pseudo}^2 , β via (5)
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$$\ell_R(\delta) = \frac{1}{2} \left[p \left(\log \frac{p}{2\pi} - 1 - \log \text{RSS} \right) - \left(\sum_i \log(\lambda_i + \delta) + \log |A| - \log |\mathbf{X}^\top \mathbf{X}| \right) \right], \quad (3)$$

with the derivative in closed form through $A' = -W^\top D^2 W$, $b' = -W^\top D^2 w$. The two formulations agree to 5×10^{-13} on likelihoods in the test suite; the lemma form replaces the second $O(n^3)$ eigendecomposition with $O(nq^2)$ work.

2.2 Profile likelihood and grid

The REML profile log-likelihood as a function of δ is The ML variant replaces p by n and drops the $\log |A| - \log |\mathbf{X}^\top \mathbf{X}|$ terms (same quadratic form). The equivalence with the restricted-eigenspace form above (equation label kept for cross-reference) is asserted in the tests; v1's deliberate factor-of-two in the derivative function – harmless for root finding – is normalized. The grid $\delta_j = \exp(\text{linspace}(l_{\min}, l_{\max}, G+1))$ with $G = 100$ and $[l_{\min}, l_{\max}] = [-10, 10]$ is evaluated *vectorized over all grid points simultaneously* by broadcasting λ^R against δ , so the whole grid costs $O(pG)$ fused operations. The maximizing bracket is located by a sign change of $d\ell/d\delta$ and polished with Brent's method (v1 used Newton from the bracket midpoint; the root is identical and Brent cannot diverge), with v1's boundary-acceptance and grid-argmax fallback rules retained. Finally

$$\hat{\sigma}_g^2 = \frac{s_1(\hat{\delta})}{p}, \quad \hat{\sigma}_e^2 = \hat{\delta} \hat{\sigma}_g^2, \quad h_{\text{pseudo}}^2 = \frac{1}{1 + \hat{\delta}}. \quad (4)$$

2.3 GLS coefficients and gBLUP

With $\mathbf{H} \equiv \mathbf{V}^{-1/2}$ (applied implicitly, Section 5),

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y} \quad \text{computed as} \quad ((\mathbf{H}\mathbf{X})^\top (\mathbf{H}\mathbf{X}))^{-1} (\mathbf{H}\mathbf{X})^\top (\mathbf{H}\mathbf{y}), \quad (5)$$

and the gBLUP of breeding values is $\hat{u} = \hat{\sigma}_g^2 \mathbf{K} \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\hat{\beta})$, with $\mathbf{V}^{-1}$ applied through the eigenbasis.

3 The association scan: EMMA as batched GEMM

3.1 Single-SNP statistic

For a candidate SNP $g \in \mathbb{R}^n$ (missing calls mean-imputed), v1 regressed the transformed, covariate-residualized phenotype on the transformed, covariate-residualized SNP by an explicit least-squares solve *per SNP*. The identical statistic has a closed form. Define, at the fitted $\hat{\delta}$,

$$\mathbf{Q} \leftarrow \text{qr}(\mathbf{H}\mathbf{X}), \quad r \leftarrow (I - \mathbf{Q}\mathbf{Q}^\top) \mathbf{H}\mathbf{y}, \quad \tilde{g} \leftarrow (I - \mathbf{Q}\mathbf{Q}^\top) \mathbf{H}g, \quad \text{RSS}_0 \leftarrow \|r\|^2. \quad (6)$$

Algorithm 2 Batched EMMAX scan

Require: fitted $\hat{\delta}$; covariates $\mathbf{X}$; phenotype $\mathbf{y}$; genotype store returning blocks $\mathbf{G} \in \mathbb{R}^{k \times n}$; block size k , dtype

- 1: $\mathbf{Q} \leftarrow \text{qr}(\mathbf{H}\mathbf{X})$; $r \leftarrow (I - \mathbf{Q}\mathbf{Q}^\top)\mathbf{H}\mathbf{y}$; $\text{RSS}_0 \leftarrow \|r\|^2$; $\nu \leftarrow n - q - 1$
- 2: **for all** blocks $\mathbf{G}$ from the store **do**
- 3: $\tilde{\mathbf{G}}^\top \leftarrow (I - \mathbf{Q}\mathbf{Q}^\top)\mathbf{H}\mathbf{G}^\top$ $\triangleright$ 2 GEMMs vs. eigenbasis + 1 small GEMM
- 4: $\mathbf{num} \leftarrow \tilde{\mathbf{G}}^\top r$; $\mathbf{den} \leftarrow \text{rowsums}(\tilde{\mathbf{G}} \circ \tilde{\mathbf{G}})$
- 5: $t^2 \leftarrow \mathbf{num}^2 / \max(\mathbf{den}, \epsilon)$; $\mathbf{RSS} \leftarrow \max(\text{RSS}_0 - t^2, 0)$
- 6: $F \leftarrow t^2 / \mathbf{RSS} \cdot \nu$; $p \leftarrow F_{1,\nu}(F)$ $\triangleright$ batched survival function
- 7: **end for**
- 8: **return** $\{p, F, \text{RSS}, \hat{\beta}_g, \text{se}(\hat{\beta}_g)\}$

The one-degree-of-freedom F statistic is the rank-one update

$$\text{RSS}(g) = \text{RSS}_0 - \frac{(\tilde{g}^\top r)^2}{\tilde{g}^\top \tilde{g}}, \quad F(g) = \left(\frac{\text{RSS}_0}{\text{RSS}(g)} - 1 \right) (n - q - 1), \quad p(g) = F_{1, n-q-1}(F(g)), \quad (7)$$

which equals v1's per-SNP regression exactly (Frobenius–Waugh–Lovell partialling out). Effect sizes and standard errors come in closed form,

$$\hat{\beta}_g = \frac{\tilde{g}^\top r}{\tilde{g}^\top \tilde{g}}, \quad \text{se}(\hat{\beta}_g) = \sqrt{\frac{\text{RSS}(g)}{(n - q - 1) \tilde{g}^\top \tilde{g}}}. \quad (8)$$

3.2 Batching

All per-SNP quantities are rows of matrix expressions. For a block $\mathbf{G} \in \mathbb{R}^{k \times n}$ of k SNPs:

$$\tilde{\mathbf{G}}^\top = (I - \mathbf{Q}\mathbf{Q}^\top)\mathbf{H}\mathbf{G}^\top, \quad \mathbf{num} = \tilde{\mathbf{G}}^\top r \in \mathbb{R}^k, \quad \mathbf{den} = \text{diag}(\tilde{\mathbf{G}}^\top \tilde{\mathbf{G}}) \in \mathbb{R}^k, \quad (9)$$

and (7) is applied elementwise. The transform $\mathbf{H}\mathbf{G}^\top$ is two GEMMs against the eigenbasis (one GEMM per side); the residualization is one small GEMM of cost $O(nqk)$. The scan is therefore $O(n^2m)$ dense linear algebra with BLAS-3 kernels, no Python-level per-SNP loop, and no per-SNP factorization; arithmetic defaults to `float32` (fitting stays `float64`).

3.3 Multi-df extensions

2-df genotypic tests (v1's ANOVA path) build $L-1$ indicator contrasts for an L -level SNP and solve the small normal equations.

GxE. The 1-df interaction test is for $c_j = 0$ in $\mathbf{y} = \mathbf{X}\mathbf{b} + g_j a_j + (g_j \circ E)c_j + u + e$, with the SNP main effect a_j in the model and E among the covariates (it is added when absent): an interaction column tested without both main effects absorbs them. With $\tilde{g} = (I - \mathbf{Q}\mathbf{Q}^\top)\mathbf{H}g$ and $\tilde{c} = (I - \mathbf{Q}\mathbf{Q}^\top)\mathbf{H}(g \circ E)$, the interaction sum of squares after the main effect is $(\tilde{c}^\top r - \frac{\tilde{c}^\top \tilde{g}}{\tilde{g}^\top \tilde{g}} \tilde{g}^\top r)^2 / (\tilde{c}^\top \tilde{c} - \frac{(\tilde{c}^\top \tilde{g})^2}{\tilde{g}^\top \tilde{g}})$, batched over the block; the 2-df joint test adds the main-effect reduction. An optional polygenic interaction component with kinship $\mathbf{K} \circ (EE^\top)$ (Sul et al. 2016) is fitted as a two-kinship mixture (Section 10). The 2026-10-02 revision omitted a_j ; with a 0/1 environment a pure main-effect SNP then gave interaction $p \approx 10^{-6}$ in 20 of 20 replicates.

MLMM. The stepwise multi-locus mixed model (Segura et al. 2012) adds the most significant SNP as a fixed-effect cofactor, re-estimates the variance components (REML for the scans) and repeats until the pseudo-heritability has vanished or a step cap is reached, then eliminates backward from the last forward model, dropping the least significant cofactor (conditional F test) each step. Every visited model is scored with the *maximum-likelihood* log-likelihood ℓ (REML likelihoods are not comparable across fixed-effect designs) by the extended BIC (Chen & Chen 2008), $\text{EBIC} = -2\ell + k' \log n + 2 \log \binom{m}{k}$, and by mBonf, the largest model whose cofactors all have conditional $p < \alpha/m$. Plain BIC is reported but, as Segura et al. found, it is too tolerant for GWAS.

4 Large n : the kinship as a streaming operator

Following BOLT-LMM, above the dense regime the kinship is never materialized. **GenotypeKinship** wraps the genotype store and applies K only through streaming products,

$$KX = \frac{1}{m} \sum_{\text{blocks}} Z_b (Z_b^\top X), \quad (10)$$

with Z_b the Yang-2010 called-only standardized block; the **scale_k** normalization constants (mean off-diagonal, mean diagonal) are obtained from two cheap streaming passes and folded into the same products, so the operator equals the dense GRM to machine precision (verified to 10^{-14} in the tests). Every large- n computation is operator-native: the randomized eigensolver, the SLQ solver of Section 6, the scan transform (through the truncated basis below) and gBLUP all consume mat-vec products only. Cost per product is $O(nm)$ tall-skinny GEMMs with $O(\text{block})$ working memory, replacing $O(n^2)$ storage and an $O(n^2m)$ materialization; the equivalence to the dense pipeline (variance components, likelihood, gBLUP, scan under gentle truncation) is tested directly.

Adaptive deflation. The SLQ solver deflates only *well-separated* extreme eigenvalues (Ritz values at least $4\times$ the bulk median): the analytic accounting for deflated directions is exact only for an invariant subspace, and subspace iteration cannot converge the boundary of a near-degenerate bulk. Spectra without outliers deflate nothing – the tight bulk is precisely the regime Gauss quadrature resolves well.

5 Large n : truncated randomized spectrum

This approximation is mixmogam’s own (BOLT-LMM never truncates the spectrum). A randomized top- k approximation ($k = 1024$ by default) is obtained from a Gaussian sketch $\Omega \in \mathbb{R}^{n \times (k+p)}$ ($p = 12$ oversampling) powered by $\ell = 5$ subspace iterations, $\mathbf{Q}_0 \leftarrow \text{orth}(\mathbf{K}\Omega)$, $\mathbf{Q}_t \leftarrow \text{orth}(\mathbf{K}\mathbf{Q}_{t-1})$, and the Ritz extraction $B = \mathbf{Q}^\top \mathbf{K} \mathbf{Q}$, $B = W \Lambda W^\top$ yields $\hat{\Lambda}_k, \hat{U}_k = \mathbf{Q} W_k$ (Halko et al. 2011). With the discarded spectrum treated as a flat bulk at its mean $\bar{\lambda} = (\text{tr } \mathbf{K} - \sum_{i \leq k} \hat{\lambda}_i)/(n - k)$,

$$\mathbf{H} \mathbf{A} = (\bar{\lambda} + \delta)^{-1/2} \mathbf{A} + \hat{U}_k \left[\text{diag}((\hat{\lambda}_i + \delta)^{-1/2}) - (\bar{\lambda} + \delta)^{-1/2} I_k \right] \hat{U}_k^\top \mathbf{A}, \quad (11)$$

exact when the discarded eigenvalues are equal. They are not: a GRM bulk has a Marchenko–Pastur spread, and the per-SNP whitening error it leaves inflates the scan. Re-read with the corrected λ_{GC} (Section 12), the 2026-10-03 large- n simulation gave $\lambda_{GC} = 2.13, 1.28$ and 0.95 at $k/n = 0.03, 0.13, 0.26$ ($n = 4,000$; exact scan 0.91) and 1.30 at $k/n = 0.10$ ($n = 10,000$; exact scan 0.85 ; 84 vs 50 significant non-causal SNPs). Large- n association therefore goes through the two-step statistics of Section 7.

Algorithm 3 Deflated SLQ variance-component fit

Require: operator $\text{mv}(x) = \mathbf{S}\mathbf{K}\mathbf{S}x$; $\mathbf{y}$, $\mathbf{X}$; probes P , steps t , deflation d

- 1: $\hat{\lambda}, \hat{U} \leftarrow$ randomized top- d of mv
 - 2: $\text{mv}_d(x) \leftarrow \text{mv}(x) - \hat{U}(\hat{\lambda} \circ (\hat{U}^\top x))$
 - 3: $y_\perp \leftarrow \mathbf{S}\mathbf{y}$; decompose $y_\perp = \hat{U}c + y_{\text{rest}}$
 - 4: rules: one Lanczos($\text{mv}_d, y_{\text{rest}}, t$); P Lanczos($\text{mv}_d, \mathbf{S}z, t$)
 - 5: **for** δ on grid and within Brent bracket **do**
 - 6: $s_1(\delta) \leftarrow \sum_i c_i^2 / (\hat{\lambda}_i + \delta) + \mathcal{Q}_y[f_\delta]$
 - 7: $s_2(\delta) \leftarrow \sum_i \log(\hat{\lambda}_i + \delta) + \overline{\mathcal{Q}}_P[\log] - d \log \delta$ $\triangleright$ deflation correction
 - 8: $s_4(\delta) \leftarrow \sum_i (\hat{\lambda}_i + \delta)^{-1} + \overline{\mathcal{Q}}_P[f_\delta] - d/\delta$
 - 9: assemble $\ell_R(\delta), \ell'_R(\delta)$ as in (??)
 - 10: **end for**
 - 11: **return** $\hat{\delta}$ by bracketing + Brent; $\hat{\sigma}_g^2 = s_1(\hat{\delta})/p$
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6 Large- n fitting: deflated stochastic Lanczos quadrature

Fitting δ without the full spectrum requires the spectral sums s_1, s_2, s_3, s_4 of Section 2. The SLQ solver estimates them from Gauss quadrature rules built by Lanczos iteration (Ubaru et al. 2017): with T_t the tridiagonal Lanczos matrix after t steps from start vector v ,

$$v^\top f(\mathcal{A}) v \approx \|v\|^2 \sum_{i=1}^t w_i f(\theta_i), \quad (\theta, w) = \text{eigenpairs of } T_t, \quad w_i = (S_{1i})^2, \quad (12)$$

exact for polynomials of degree $\leq 2t-1$ and spectrally accurate for analytic f such as $f_\delta(t) = (t + \delta)^{-1}$ and $\log(t + \delta)$. Trace terms use unbiased Rademacher probe averages ($\mathbb{E}[z^\top f(\mathcal{A})z] = \text{tr } f(\mathcal{A})$), probes pre-projected into the covariate complement for REML; the quadratic forms in $\mathbf{y}$ use one deterministic rule from the projected phenotype. All Lanczos runs use full reorthogonalization ($t \leq 96$ default, 12 probes).

Deflation. Gauss quadrature converges slowly when the spectrum mixes extreme outliers (population structure) with a tight bulk. Extreme eigenpairs of the operator are therefore estimated by a randomized top- d pass ($d = 128$) and handled analytically; the quadrature resolves only the deflated bulk. Because the deflated operator carries the removed directions as (near-)zero eigenvalues, the probe estimates must be corrected by their contribution $f_\delta(0)$:

$$\sum_{i>p-d} f(\lambda_i) = \left[\text{probe mean of } f_\delta\text{-rules} \right] - d f_\delta(0), \quad f_\delta(0) \in \{1/\delta, \log \delta, \delta^{-2}\}. \quad (13)$$

Quadrature nodes are floored at zero to guard against roundoff-negative deflated eigenvalues. On structure-rich kinships the resulting $\hat{\delta}$ agrees with the exact EMMA fit to a few percent (the REML profile is very flat near its optimum) and profile likelihoods agree to about one nat; the total cost is $O(n^2)$ per matvec times $O(\text{probes} \times \text{steps})$ matvecs, with no $O(n^3)$ factorization.

7 LOCO association and two-step statistics

Why LOCO. With the tested SNP inside the kinship, the polygenic term absorbs part of its own effect (proximal contamination; Listgarten et al. 2012, Yang et al. 2014): the exact EMMAX

scan of the 2026-10-03 simulation was deflated ($\lambda_{GC} = 0.91$ at $n = 4,000$, 0.85 at $n = 10,000$). Every method behind `gwas()` therefore tests SNPs on chromosome g against a model built from the other chromosomes. With more than 25 chromosomes (e.g. simulated data with one “chromosome” per LD block) consecutive chromosomes are merged into 25 groups of balanced size, BOLT-LMM’s genome segments.

Exact LOCO. For each group g , $\mathbf{K}_{-g} = \text{scale}((\sum_j z_j z_j^\top - \sum_{j \in g} z_j z_j^\top)/(m - m_g))$ is formed by subtraction (two passes over the genotypes in total), the variance components are refitted by exact EMMA, and the SNPs of group g are scanned with Algorithm 2. Cost: one $O(n^3)$ eigendecomposition per group.

The streaming LOCO operator. The two-step methods hold the genotypes as standardized, covariate-projected blocks Z_b ($z \leftarrow z - \mathbf{Q}\mathbf{Q}^\top z$, BOLT-LMM’s treatment of covariates) that never straddle a group. One pass applies *every* LOCO kinship to its own column of a block P : with per-group products $S_h = Z_h^\top (Z_h P)$,

$$\mathbf{K}_{-g} P_{:,r} = \frac{(\sum_h S_h - S_g)_{:,r}}{M - M_g}, \quad g = \text{group}(r), \quad (14)$$

so the G LOCO systems of BOLT-LMM, the calibration solves, and the LOCO eigenbases below all share GEMM passes.

7.1 BOLT-LMM-inf

Variance components (σ_g^2, δ) are estimated once on all SNPs (Section 6; BOLT-LMM uses Monte Carlo REML for the same quantities) and are not refitted per group, BOLT-LMM’s default. For each group, $w_g = (\mathbf{K}_{-g} + \delta I)^{-1} \mathbf{y}$ is solved by batched preconditioned conjugate gradients, the preconditioner inverting the top-64 eigenpairs of $\mathbf{K}$ exactly and the rest as a flat bulk. Every SNP z_j in group g then gets the retrospective statistic

$$\chi_j^2 = \frac{(z_j^\top w_g)^2}{c(z_j^\top z_j / n_{\text{eff}}) \|w_g\|^2}, \quad n_{\text{eff}} = n - q, \quad (15)$$

whose calibration constant c makes it match the exact prospective statistic $(z^\top \mathbf{V}_{-g}^{-1} \mathbf{y})^2 / (z^\top \mathbf{V}_{-g}^{-1} z)$: for 30 random SNPs with GRAMMAR $\chi^2 < 5$, $c_j = \sigma_g^2 z_j^\top (\mathbf{K}_{-g} + \delta I)^{-1} z_j / ((z_j^\top z_j / n_{\text{eff}}) \|w_g\|^2)$ by 30 more CG solves, and $c = \bar{c}_j$ (Loh et al. 2015). The spread $\text{cv} = \text{sd}(c_j)/c$ is reported: (15) is exact only if the prospective denominator is proportional to $z_j^\top z_j$ across SNPs, an assumption the BOLT-LMM authors flagged as untested outside human data.

7.2 BOLT-LMM

The prior on standardized SNP effects is the two-Gaussian mixture

$$\beta_j \sim p \mathcal{N}\left(0, \frac{(1-f_2)\sigma_\beta^2}{p}\right) + (1-p) \mathcal{N}\left(0, \frac{f_2\sigma_\beta^2}{1-p}\right), \quad \sigma_\beta^2 = \sigma_g^2/M, \quad (16)$$

fitted by fully factorized variational Bayes (iterated conditional posterior means). (f_2, p) is chosen on BOLT-LMM’s 18-point grid $f_2 \in \{0.5, 0.3, 0.1\}$, $p \in \{0.5, 0.2, 0.1, 0.05, 0.02, 0.01\}$ by 5-fold cross-validated prediction; the mixture is used only if its CV R^2 beats the infinitesimal $(0.5, 0.5)$ by a

relative 1% (BOLT-LMM’s threshold is not published; this is our choice), otherwise BOLT-LMM-inf is returned. LOCO residuals $r_g = \mathbf{y} - Z_{-g}^\top \hat{\beta}_{-g}$ replace w_g in (15), and the constant is set so that the LD Score regression intercept (Bulik-Sullivan et al. 2015) of the mixture statistic equals that of BOLT-LMM-inf. In-sample LD scores use bias-adjusted r^2 within 1 Mb. When the LD scores barely vary (coefficient of variation < 0.2 , e.g. LD-free simulations) the intercept is not identified and the medians are matched instead.

Blocked variational Bayes. All cross-validation fits (18 settings $\times$ 5 folds = 90 columns) or all LOCO fits (G columns) run as one Gauss–Seidel iteration over SNP blocks of 128: the block’s products $Z_b^\top R$ are one GEMM, and as SNP a moves by Δ_a the products of later SNPs in the block are corrected through the block’s Gram matrix, $u_c \leftarrow u_c - G_{ca} \Delta_a$ (per-fold Gram matrices for cross-validation columns). The sequential part is a $B \times B \times$ columns loop (Numba), the rest is BLAS. The posterior mean of one effect given the others is closed-form for both priors used here (the Laplace component via truncated normals).

7.3 LDAK-KVIK

Following the LDAK-KVIK documentation (Hof & Speed 2025): (1) structure test: among 512 SNPs spread over the genome, the mean squared correlation between SNPs on different chromosomes, $\text{excess} = n_{\text{eff}} \bar{r}^2 - 1$; strong structure if $\text{excess} > 0.1$ with $z > 3$; (2) per-SNP heritabilities $h_j^2 = w_j h^2 / \sum_k w_k$ with $w_j = [f_j(1-f_j)]^{1+\alpha}$ and α the best of $\{-1, -0.75, -0.5, -0.25, 0\}$ by REML likelihood of the weighted kinship (LDAK-KVIK: randomized Haseman–Elston regression; no LD thinning here); (3) the elastic-net prior $\gamma_j \sim p \text{Laplace}(\sqrt{2p/((1-F)h_j^2)}) + (1-p)\mathcal{N}(0, Fh_j^2/(1-p))$ with (p, F) chosen on a 10% hold-out among KVIK’s ten pairs; (4) LOCO polygenic scores P_g by variational Bayes; (5) the ordinary least-squares statistic of $\mathbf{y} - P_g$ on z_j , scaled by λ : $\lambda = 1$ without strong structure, otherwise $\lambda = \lambda' \sum_S T / \sum_S U$ over SNPs with $\lambda U < 6$ and $\lambda' T < 6$, where T are the statistics with ridge-prior scores and $\lambda' = 1/c$ their GRAMMAR-Gamma calibration (the BOLT-LMM-inf constant of the weighted model).

7.4 Structure-aware denominator

Every statistic above, and REGENIE’s and SAIGE’s, scales a score by one genome-wide constant. Under strong structure the prospective denominator $z_j^\top (\mathbf{K}_{-g} + \delta I)^{-1} z_j$ is *not* proportional to $z_j^\top z_j$: a SNP aligned with the leading (structure) eigenvectors of $\mathbf{K}_{-g}$ has a smaller denominator than an unaligned SNP of equal norm, so one constant inflates unaligned SNPs and deflates aligned ones (Section 12). mixmogam’s extension replaces the constant by a low-rank-plus-flat approximation on the top- k eigenpairs (U_g, Λ_g) of the SNP’s own LOCO kinship,

$$D_j = c'_g \sigma_g^2 \left[\|(\Lambda_g + \delta)^{-1/2} U_g^\top z_j\|^2 + \frac{z_j^\top z_j - \|U_g^\top z_j\|^2}{\bar{\lambda}_g + \delta} \right], \quad \chi_j^2 = \frac{(z_j^\top w_g)^2}{D_j}, \quad (17)$$

with $\bar{\lambda}_g$ the mean of the remaining eigenvalues of $\mathbf{K}_{-g}$ and c' calibrated on the same 30 prospective SNPs. The width k is adaptive: it doubles from 64 to at most 512 until the spread of c'_j over the calibration SNPs falls below 3%. The bulk of an LD-rich kinship is not flat, and the calibration SNPs measure how much of it the basis misses (in a coalescent sample without population structure, $n = 1,500$, $k = 64$ left a 7% spread and $k = 256$ 2%). The LOCO eigenbases are computed for all groups at once by randomized subspace iteration through (14). A basis of the *full* kinship does not work: it contains each SNP’s own LD block and reintroduces proximal contamination into the

Algorithm 4 BOLT-LMM-inf with the structure-aware denominator

Require: standardized projected blocks Z_b in G LOCO groups; $\mathbf{y}$; (σ_g^2, δ) ; width k

- 1: $W \leftarrow$ batched PCG for $(\mathbf{K}_{-g} + \delta I) w_g = \mathbf{y}$, $g = 1..G$ $\triangleright$ one LOCO pass per iteration, (14)
- 2: $\{U_g, \Lambda_g, \bar{\lambda}_g\} \leftarrow$ batched randomized subspace iteration on all $\mathbf{K}_{-g}$
- 3: 30 SNPs j with GRAMMAR $\chi^2 < 5$: $v_j \leftarrow (\mathbf{K}_{-g(j)} + \delta I)^{-1} z_j$ $\triangleright$ batched PCG
- 4: $c' \leftarrow \text{mean}_j (\sigma_g^2 z_j^\top v_j / D_j^{(c'=1)})$
- 5: **for all** blocks Z_b of group g **do**
- 6: **num** $\leftarrow Z_b w_g$; $P \leftarrow Z_b U_g$; $D \leftarrow c' \sigma_g^2 [(P \circ P)(\Lambda_g + \delta)^{-1} + (\text{rowsums}(Z_b \circ Z_b) - \text{rowsums}(P \circ P)) / (\bar{\lambda}_g + \delta)]$
- 7: $\chi^2 \leftarrow \text{num}^2 / D$
- 8: **end for**

denominator. For the mixture and elastic-net statistics, the per-SNP factor $\kappa_j = D_j^{\text{const}} / D_j$ of the infinitesimal model is applied before their global calibration; this transfer is a heuristic.

8 Kinships

Additive GRM. Per variant j , genotypes are standardized over *called* samples only, $z_{ij} = (g_{ij} - \bar{g}_j) / s_j$ for called g_{ij} and $z_{ij} = 0$ otherwise (Yang et al. 2010), and $\mathbf{K} = \sum_j w_j z_j z_j^\top / \sum_j w_j$ is accumulated over SNP blocks in `float64`. Optional weights w_j implement LD&K-style weighting; an optional SNP subset fits the null model from a marker subsample at reduced cost.

IBS. Identity-by-state similarity is accumulated as one-hot GEMMs per block: with H_a the indicator of genotype $a \in \{0, 1, 2\}$ over called entries, the shared-allele numerator is $\sum_a H_a H_a^\top$ and the denominator is $CC^\top$ with C the called-indicator, so $\mathbf{K}_{\text{IBS}} = S/C$ per pair.

Exact LOCO. Because the GRM is a sum of globally standardized outer products, the blocks accumulated for each chromosome $\mathbf{K}_c = m^{-1} \sum_{j \in c} z_j z_j^\top$ satisfy $\sum_c \mathbf{K}_c = \mathbf{K}$ *exactly* under a single global standardization, and the leave-one-chromosome-out matrix is the algebraic difference

$$\mathbf{K}_{\text{loco}}(c) = \frac{m \mathbf{K} - m_c \mathbf{K}_c}{m - m_c}, \quad (18)$$

rescaled to mean diagonal one. No per-chromosome restandardization is performed, hence no approximation. Windowed local/global pairs along the genome are the same construction restricted to a sliding interval.

9 Batched permutation thresholds

Permuting the raw phenotype breaks the covariance the kinship models: under relatedness or structure the permuted phenotypes are not draws from the null model, and the threshold is anti-conservative (Abney 2015). With 4 populations at $F_{ST} = 0.3$, the v1 scheme gave a family-wise error rate of $7.6\% \pm 0.6\%$ at a nominal 5% (2,000 parametric-null replicates, 1,000 permutations). mixmogam therefore permutes the GLS-whitened null residuals (Abney’s MVNpermute construction): $r = (I - \mathbf{Q}\mathbf{Q}^\top) \mathbf{H}\mathbf{y}$ at the fitted $\hat{\delta}$ is close to exchangeable under the null, so

$$R_P = (I - \mathbf{Q}\mathbf{Q}^\top) r[\Pi] \in \mathbb{R}^{n \times B} \quad (19)$$

holds B draws of the whitened null. All permutations then flow through each SNP-block GEMM at once,

$$T = \tilde{\mathbf{G}}^\top R_P \in \mathbb{R}^{k \times B}, \quad F = \frac{T \circ T}{\text{den RSS}_{0,P}} \nu, \quad (20)$$

and the genome-wide 5% threshold is the 5th percentile of the per-permutation minimum p . The v1 scheme remains available (`scheme="raw"`) to reproduce old results.

10 Two-kinship mixtures

For $\mathbf{V} = \sigma_{g1}^2 \mathbf{K}_1 + \sigma_{g2}^2 \mathbf{K}_2 + \sigma_e^2 I$, the ratio $\rho = \sigma_{g1}^2 / \sigma_{g2}^2$ is profiled on a grid of $\log \rho$ over $[-3, 3]$ (cascading refinements around the best bracket), fitting each candidate mixture $a\mathbf{K}_1 + (1-a)\mathbf{K}_2$ by the exact EMMA path of Section 2; variance shares are reported per matrix.

11 Missing data

Per-SNP sample subsetting (v1’s behavior) breaks block batching. The default policy is a per-variant missingness filter followed by mean imputation over called genotypes (EMMAX-consistent), fused into the int8-to-float block conversion (optionally a Numba kernel); `impute='none'` propagates NaN for analyses that prefer explicit handling.

12 Validation and measured performance

Correctness. The exact scan is certified against a float64 port of the v1 mathematics, with F statistics equal to $\text{rtol } 10^{-6}$ (the oracle is `tests/_reference.py`). The exact LOCO path equals a hand-built per-chromosome EMMAX fit; the streaming LOCO operator equals the dense LOCO kinship products; batched CG equals a dense solve; variational Bayes with a Gaussian prior converges to the BLUP residual $\sigma_e^2 \mathbf{V}^{-1} \mathbf{y}$; both posterior means match numerical quadrature; LD scores match a brute-force computation; whitened-residual permutations reproduce scans of $\mathbf{V}^{1/2} P r$ and keep the family-wise error at its nominal level under strong structure; the GxE statistic equals a direct GLS F test of the full design. Without structure, BOLT-LMM-inf agrees with the exact LOCO scan (chi-square correlation 0.996–0.9998, mean ratio 0.99–1.00), and mixmogam’s LDAK-KVIK correlates 0.966 per SNP with the LDAK 6 binary.

The constant-ratio assumption under structure. Table 1 uses a null-chromosome design: the genetic value comes from all chromosomes but the last (polygenic $h^2 = 0.4$ on standardized genotypes plus 10 QTLs with $h^2 = 0.1$), so the last chromosome’s SNPs are null and their LOCO kinship is the generating one. SNPs are binned by their squared-norm share in the top-10 kinship eigenvectors. Data: LD-free simulated genotypes ($n = 1,300$, $m = 20,000$; 4 populations at $F_{ST} = 0.3$) and the *A. thaliana* RegMap genotypes ($n = 1,307$ inbred accessions, 53,167 SNPs with $\text{MAC} \geq 10$). Six replicate phenotypes each (archives 20261003T081812Z-`structure-calibration` and 20261003T083746Z-`kvik-reference`).

Without structure every method is flat (λ_{GC} 0.98–1.04). Under structure, each single-constant statistic is anti-conservative for SNPs that load weakly on the structure and conservative for strongly loading ones, while its genome-wide λ_{GC} (0.97–1.04) looks normal. The structure-aware denominator restores per-quintile calibration for BOLT-LMM-inf. Transferred to BOLT-LMM it is flat (0.96–1.09). Transferred to LDAK-KVIK through its λ rule it is flat but about 7% inflated overall

Table 1: λ_{GC} on null SNPs by structure-loading quintile (q1 lowest, q5 highest) and false-positive rate at $p < 0.01$, means over six replicates. “spectral” is Section 7.4; “LDAK binary” is the reference LDAK-KVIK program run on the same phenotypes.

data	method	q1	q2	q3	q4	q5	FPR q1 / q5
sim., $F_{ST} = 0.3$	exact LOCO EMMAX	1.05	0.98	1.05	0.92	1.02	1.12% / 1.11%
	BOLT-LMM-inf	1.34	1.12	1.07	0.82	0.64	2.45% / 0.11%
	BOLT-LMM-inf, spectral	1.05	0.99	1.05	0.92	1.01	1.14% / 1.06%
	BOLT-LMM	1.35	1.14	1.06	0.85	0.70	2.58% / 0.11%
	LDAK-KVIK (mixmogam)	1.37	1.17	1.12	0.89	0.73	2.92% / 0.11%
<i>A. thaliana</i>	exact LOCO EMMAX	1.04	1.01	0.98	1.01	1.04	0.96% / 0.98%
	LDAK-KVIK, LDAK binary	1.21	1.10	1.04	1.01	0.92	1.67% / 0.78%
	LDAK-KVIK (mixmogam)	1.16	1.07	0.99	0.98	0.85	1.44% / 0.49%
	BOLT-LMM-inf	1.19	1.10	1.01	0.99	0.86	1.66% / 0.56%
	BOLT-LMM-inf, spectral	1.04	1.02	0.98	1.00	1.03	0.97% / 0.99%

under the simulated structure. Power at the QTLs, which are not chosen for structure alignment, is within a few percent across methods.

Speed. Development laptop, Apple silicon, 4 BLAS threads, AC power; these compare mixmogam with itself.

Workload	Metric	Result
exact scan vs. v1-style per-SNP least squares ($n=2000$, $m=10k$)	speed-up	$123\times$
exact scan, $n=2000$, $m=100k$, float32	throughput	4.8×10^4 SNPs/s
gwas exact LOCO, <i>A. thaliana</i> ($n=1307$, $m=53k$, 5 chr.)	wall time	4 s
gwas BOLT-LMM-inf, same data (spectral: +2 s)	wall time	28 s
gwas LDAK-KVIK, same data (LDAK binary: 9 s)	wall time	140 s

13 References

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